



Test Project: Advanced Retail Inventory and Sales Management System

TechStack

Frontend	AngularJS
Backend	NestJS
Deployment	Docker
ORM	Typeorm
Database	mySQL

Description:

Develop an advanced inventory and sales management system for a retail store. The system should allow store employees and managers to:

Add, update, and delete products with details like name, price, stock quantity, and product category.

View a list of all products with the ability to search and filter by name, price range, or category.

Perform advanced search and filtering, e.g., full-text search, multiple filters, etc.

Process sales transactions by adding products to a cart, calculating the total price, updating stock quantities, and generating invoices.

View real-time updates of stock quantities on the product listing page.

View a list of all sales transactions with total amounts and products sold.

Generate reports on sales, stock levels, and other metrics.

Optimize the performance of the application to handle a large number of products and transactions.

Frontend (AngularJS):

- Implement all required pages and features as described in the description.
- Implement real-time updates of stock quantities on the product listing page.
- Optimize the performance of the frontend application to handle large datasets and ensure a smooth user experience.

Backend (NestJS):

- Implement all required APIs as described in the description.
- Use TypeORM to interact with the database.
- Implement real-time updates using websockets or another real-time communication method.
- Optimize the performance of the backend application to handle a large number of requests and large datasets.
- Implement authentication and authorization to restrict access to certain features.

Docker:

Containerize the frontend and backend applications using Docker to simplify the deployment process.

Rubric

Criteria	Maximum Points	Description
Code Quality and Structure	20	Clean and well-structured code (10 points) Proper use of AngularJS and NestJS best practices (10 points)

Functionality	30	Proper implementation of product CRUD operations (10 points) Proper implementation of sales transactions and stock update (10 points) Proper implementation of search and filter functionality (5 points) Proper error handling and validation (5 points)
Database Integration	20	Proper use of TypeORM for database interactions (10 points) Proper handling of data relationships and constraints (10 points)
Real-Time Updates	10	Proper implementation of real-time updates (10 points)
Performance Optimization	10	Proper optimization of the frontend and backend applications (10 points)
User Interface	10	- Intuitive and user-friendly design (5 points) - Proper implementation of responsive design (5 points)
Total	100	